KAIZEN MATHS

Lesson starter

Advanced Matrices: Transformations and Eigenvalues

Eigenvalues and Eigenvectors

Name _____

S1

Find the characteristic equation of each matrix.

$A = \begin{bmatrix} 3 & -3 \\ 0 & 7 \end{bmatrix}$

S2

Find the eigenvalues of each matrix.

$A = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$

S3

Find an eigenvector for the stated eigenvalue.

$A = \begin{bmatrix} 1 & 1 \\ 0 & 6 \end{bmatrix}, \lambda = 6$

S4

Write each matrix in diagonalised form.

$A = \begin{bmatrix} 1 & -3 \\ 0 & 6 \end{bmatrix}$

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Exit ticket

Advanced Matrices: Transformations and Eigenvalues

Eigenvalues and Eigenvectors

Name _____

E1

Find the characteristic equation of each matrix.

$$A = \begin{bmatrix} 2 & -3 \\ 0 & 6 \end{bmatrix}$$

E2

Find the eigenvalues of each matrix.

$$A = \begin{bmatrix} 2 & 2 \\ 0 & 4 \end{bmatrix}$$

E3

Find an eigenvector for the stated eigenvalue.

$$A = \begin{bmatrix} 5 & -2 \\ 0 & 7 \end{bmatrix}, \lambda = 7$$

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Eigenvalues and Eigenvectors

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Find an eigenvector for the stated eigenvalue.

$A = \begin{bmatrix} 5 & -2 \\ 0 & 7 \end{bmatrix}, \lambda = 7$

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Find an eigenvector for the stated eigenvalue.

$A = \begin{bmatrix} 5 & -2 \\ 0 & 7 \end{bmatrix}, \lambda = 7$

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Teacher answers & working

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S1

Find the characteristic equation of each matrix.

$A = \begin{bmatrix} 3 & -3 \\ 0 & 7 \end{bmatrix}$

Answer:

$\lambda^2 - 10\lambda + 21 = 0$

Working:

For a 2 by 2 matrix, the characteristic equation is det(A - λI) = 0.

For A = $\begin{bmatrix} 3 & -3 \\ 0 & 7 \end{bmatrix}$, trace(A) = 10 and det(A) = 21.

So the characteristic equation is λ^2 - (trace A)λ + det(A) = 0.

Therefore λ^2 - 10λ + 21 = 0.

S2

Find the eigenvalues of each matrix.

$A = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$

Answer:

$\lambda = 1, 2$

Working:

Find the characteristic equation.

trace(A) = 3 and det(A) = 2.

$\lambda^2 - 3\lambda + 2 = 0$.

Factorise: (λ - 1)(λ - 2) = 0.

Therefore the eigenvalues are λ = 1 and λ = 2.

Teacher answers & working

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S3

Find an eigenvector for the stated eigenvalue.

$A = \begin{bmatrix} 1 & 1 \\ 0 & 6 \end{bmatrix}, \lambda = 6$

Answer:

$\begin{bmatrix} 1 \\ 5 \end{bmatrix}$

Working:

An eigenvector v satisfies (A - λI)v = 0.

$A - 6I = \begin{bmatrix} -5 & 1 \\ 0 & 0 \end{bmatrix}$.

Let $v = \begin{bmatrix} x \\ y \end{bmatrix}$.

The first row gives -5x + y = 0.

Choose y = 5.

Then -5x + 5 = 0, so x = 1.

One eigenvector is $\begin{bmatrix} 1 \\ 5 \end{bmatrix}$.

S4

Write each matrix in diagonalised form.

$A = \begin{bmatrix} 1 & -3 \\ 0 & 6 \end{bmatrix}$

Answer:

$P = \begin{bmatrix} 1 & -3 \\ 0 & 5 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}$

Working:

Use eigenvectors as columns of P.

For λ = 1, an eigenvector is $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

For λ = 6, an eigenvector is $\begin{bmatrix} -3 \\ 5 \end{bmatrix}$.

So $P = \begin{bmatrix} 1 & -3 \\ 0 & 5 \end{bmatrix}$.

The diagonal matrix D contains the eigenvalues in the same order as the eigenvectors.

$D = \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}$.

Therefore $A = PDP^{-1}$.

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E1

Find the characteristic equation of each matrix.

$A = \begin{bmatrix} 2 & -3 \\ 0 & 6 \end{bmatrix}$

Answer:

$\lambda^2 - 8\lambda + 12 = 0$

Working:

For a 2 by 2 matrix, the characteristic equation is det(A - λI) = 0.

For $A = \begin{bmatrix} 2 & -3 \\ 0 & 6 \end{bmatrix}$, trace(A) = 8 and det(A) = 12.

So the characteristic equation is λ^2 - (trace A)λ + det(A) = 0.

Therefore λ^2 - 8λ + 12 = 0.

E2

Find the eigenvalues of each matrix.

$A = \begin{bmatrix} 2 & 2 \\ 0 & 4 \end{bmatrix}$

Answer:

$\lambda = 2, 4$

Working:

Find the characteristic equation.

trace(A) = 6 and det(A) = 8.

$\lambda^2 - 6\lambda + 8 = 0$.

Factorise: (λ - 2)(λ - 4) = 0.

Therefore the eigenvalues are λ = 2 and λ = 4.

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E3

Find an eigenvector for the stated eigenvalue.

$A = \begin{bmatrix} 5 & -2 \\ 0 & 7 \end{bmatrix}$, $\lambda = 7$

Answer:

$\begin{bmatrix} -2 \\ 2 \end{bmatrix}$

Working:

An eigenvector v satisfies (A - λI)v = 0.

$A - 7I = \begin{bmatrix} -2 & -2 \\ 0 & 0 \end{bmatrix}$.

Let $v = \begin{bmatrix} x \\ y \end{bmatrix}$.

The first row gives -2x - 2y = 0.

Choose y = 2.

Then -2x - 4 = 0, so x = -2.

One eigenvector is $\begin{bmatrix} -2 \\ 2 \end{bmatrix}$.